ANNEXURE I

FORMAT - TECHNICAL SPECIFICATION

For Supply, installation, testing and commissioning of extruder feed manufacturing units with Grinder, Mixer and extruder including accessories and electrical panel

Note: Kindly adhere to the technical specification of your firm as per the technical specifications mentioned in Annexure I. If you are adhering to the technical specifications mentioned in annexure I, then kindly mention as "as per requirement" in column I. If any deviation or made, mention the deviations very clearly in column II.

Sr.No.	Description and Required specification (Column I)	Mention your firm specifications without deviations for all parameters. Mention specifically if any deviations made (Column II)
	Extruder feed manufacturing unit comprises following items viz., grinder, mixer and extruder including accessories and electric panel for extruded feed preparation by utilizing distiller's grain with other agro-industrial by products. The entire unit should be operational in approximately 400 sq.ft area. The quality of stainless steel used should be SS316 or above and certificate on material used should be declared. S**uitable parts of the extruded feed manufacturing unit which are worn out due to wear and tear should be supplied additionally to cover three years guarantee period wherever required.**	
1.	**HAMMER MILL – 80-100 kg/hr** Full screen hammer mill has 80 % screening area. In grinding, chamber gap between tip and screen is very important for efficiency as well as quality of grinding. The hammer mill should comprise of a heavy-duty suitable steel fabricated body (alternatively stainlesssteel construction can be provided in hammer mill wherever necessary). The grinding chamber of hammer mill should be lined with serrated wear plates, which protects the body from wear and tear. A dynamically balanced rotor in hammer mill with a set of swing hammers accelerates the grinding process. The screen classifier forms the lower half of the grinding chamber of hammer mill. The hammer mill should be a versatile grinding unit of heavy and rugged construction and built for	

	continuous operation day after day. The hammer mill is especially designed for the course, and medium fine size reduction. • Rotor should be designed to rotate on both the sides so that when hammer is worn out on one side, second edge of hammer could be used by changing direction of rotation. This reduces time of changing hammers. Also hammer mill rotor will be dynamically balanced to minimize vibration. • For changing of bearings etc. rotor could be disassembled from hammer mill body easily and quickly without removing any of machines over hammer mill. • For efficient power supply motor should be installed on common base frame and directly connected to hammer mill through flexible coupling. • Hammers and hammer rods of hammer mill should have resistance to wear and tear, highly balanced for vibration free performance. • Hammer mill screens should have maximum holes per unit area to have more output and life of screen. • Optimum screening area per unit power for efficient grinding, screen changing should be very quick and easy without use of any tools. **Features:** • Capacity: 80-100 kg/h based on raw material • Motor for the hammer mill and Gear motor for the feeder with power: 5 HP • Make: Siemens/Bharat Bijli/Crompton • Voltage: Three Phase 415 V 50 Hz • No of Hammer: 4 • RPM Speed: 2600-2800 • Provisions for providing three 3mm, 5mm and 7mm in alloy with MS body • Bearing block, pulley, basement with accessories, support frame will be in Mild Steel/Cast Iron • MOC of outer frame: Mild Steel with Powder coated	

	• MS sheet Thickness: 5 mm • Raw material particle size < 10 mm • Finished product size: 400 to 500 microns (35 to 40 mesh) • Dimension: 1.7 x 1.5 x 1.7 m • Suitable electrical panel board with voltage switch control like MCB.	
2.	**BATCH MIXER –15 kg/Batch** Batch mixers produce a very homogeneous mix of different ingredients used in the feed industry. The vertical mixer is most ideal for mixing.The principle is not to stir or trumble but to whirl the ingredients in the mixing chamber. Vertical screw Conveyor lifts the ingredients to top, where mixing paddles whirls particles to the wall of mixing chamber following to the bottom. This mix is lifted and whirled again & again to give homogeneous mix. The biggest advantage of this type mixer is that both feeding and discharge are at ground level. In this machine, mixing is performed by paddles which revolve to give continuous turbulent movement to the ingredients. Mixers should be fitted with well-designed oil immersed gearbox with helical teeth heats treated gears which ensures prolonged trouble-free service. Mixing Paddles should be specially designed for efficient mixing and have resistance to bend under various loads. For easy operation and saving of one worker, we have the option to provide handle close to an outlet to operate it for bagging. It should mix all the ingredients uniformly. **Features:** • Capacity: 15 kg/Batch • Optimum mixing time: 6-8 min per batch • Motor Power: 2.0 HP • Make: Siemens/Bharat Bijli/Crompton • RPM:310 • MOC of Drum: Stainless Steel • SS sheet Thickness: 1.5 mm • Mixing Paddles: Two	

	• Mutually perpendicular arranged paddles • MOC of door and Angle: Mild steel with Powder coated • Door sheet Thickness: 1 mm Thickness • Nylon wheel for easy handling and movement • Moisture adding funnel included on top of the drum • Mixed material discharge window available • ON/OFF switch • Dimension(m): 1.0 ×0.5 ×1.1 • Mixer should have coefficient variation of 5 to 6 %. • Mixer outlet type and the position of outlet in the mixer should be mentioned. It should ensure complete emptying of mixer • Suitable electrical panel board with voltage switch control like MCB.	
3.	• **PRE-CONDITIONER – up to 100 kg/hr** The extruding process require preliminary treatment of raw materials before going through the twin-screw extruder. The preconditioner is used mainly for cooking the feed ingredients. The preconditioner heats, hydrates and mixes the dry raw materials before extrusion and automatically doses the pre-treated product into the extruder. It optimizes the capacities and the performance of the extruder and enhances final product features.The preconditioner carries out a continuous mix upstream, thus optimizing the processing of the product in the extruder. It ensures optimal conversion of starches, destroys certain enzymes and micro-organisms to ensure food safety and taste features. It also minimizes wear and production breakdowns since preconditioned products are perfectly homogenous and ensure consistent recipes. **Features:** • Capacity: up to 100 kg/hr • Motor Power: 1 H • Make: Siemens/Bharat Bijli/Crompton • Voltage: Three Phase 415V 50 Hz	

	• Pipe Diameter: min 80 mm • Pipe Length: min 4 ft • Steam and water can be added with raw materials as per requirement • Steam input point • Mixing speed and holding time shall be adjusted using VFD; • MOC: Stainless Steel 304; • Steam consumption: 15-20 kg/hr • Dimension(m): 1.3 x 0.4 x 1.0 approx. • Suitable electrical panel board with voltage switch control like MCB.	
4.	**TWIN SCREW EXTRUDER – 80-100 kg/hr,** A twin screw extruder is a machine consisting of two intermeshing, co-rotating screws mounted on check shafts in a closed barrel. The screws are tight and self-wiping, which eliminates stagnant zones over the entire length of the process section. This results in high efficiency and perfect self-cleaning. Twin screw extruders can be used in feed industries for mixing a variety of feed ingredients. Twin screw extruders are able to process a large range of raw materials consistently with high levels of flexibility and pumping efficiency. The intermeshing screws allow handling of viscous, oily, sticky or very wet materials and provide a very intense mixing, where macromixing and micromixing result in a very homogeneous melt with excellent lipid binding. In a co-rotating twin screw extruder, throughput and screw speed are not interdependent; for a given formulated feed mix, the multiple operating points combine with a high control efficiency of the barrel temperature and consequently ensures efficient control of the expansion of the melt at the die. It is utilized for cooking, puffing and extrusion of raw materials into animal food of different size. **Features:** • Capacity: 80-100 kg/h, • Product type: cattle feed • Dimension (m): 2.6 ×0.9 ×1.6	

	• Motor Make: Siemens/Bharat Bijli/Crompton • Drives Make: Danfoss/Delta/Crompton Greaves • Extruder Frame: Extruder frame of mild steel construction. • MOC of outer covers: Mild steel • Extruder Drive: Gear box Specially designed gear box with oil lubrication arrangement with a separate oil pump motor. • Oil Pump Power: 0.37 Kw • Extruder Main Motor: 22 kw drive motor with ac variable frequency drive; speed ofmotor 1440 rpm; coupling connection to gear box • Suitable electrical panel board with voltage switch control like MCB.	
4a.	**Extruded Barrel** No of Barrel: 2 Nos with individual heating arrangement for each barrel, with minimum Three electrical heating elements are provided. Power consumed per heater is 2kW. Temperature: Ambient to 250 °C • End barrel is jacketed with solenoid valve arrangement which allows flow of water as per requirement to control temperature, MOC: Nitriding steel with auto Self-cleaning function without disassembling the Barrels.	
4b.	**Twin screw** • Screw Length: 1050 mm, Screw diameter: 65 mm, Screw speed: 400 RPM • Co-rotating, fully intermeshing, self-cleaning extruder screws • Screw material: High grade alloy with Nitriding treatment. There are two coupling stick between extruder barrel and gear box, which is protected by the housing coupling stick	
4c.	**Raw material feeder** Volumetric feed screw, min 30 L hopper bin driven by 0.37 kw motor with ac variable frequency drive for feeder screw speed adjustment; max screw speed is 50 to 100 rpm and Stainless-steel product contact surfaces.	

4d.	**Die Head & extruded Die** • Die mold made of special alloy will be provided with extruder for making animal feed. Three dies (3mm, 5mm and 7 mm) will be given free of cost as per customer choice.	
4e.	**Cutter Assembly** • Two to four cutter blades can be used in blade stand at a time with variable cutting speed, Cutter blade is made of heat-treated stainless steel driven by0.75kw drive motor with ac variable frequency drive. Cutter assembly should be covered with stainless steel hood • Control Panel: Start/stop push buttons available and completely Rat Proof Design. Water tight mild steel enclosure with digital indicator of feed screw speed, temperature in barrel, extruder screw speed, cutter speed etc.	
5.	**STEAM BOILER –15-20 kg/ hr** A steam boiler is a closed vessel, generally made of steel, in which water is heated by some source of heat produced by combustion of fuel and ultimately to generate steam. The steam produced may be supplied at low pressure for cooking the feed ingredients and for producing hot water which can be used for heating installations at much low pressure. **Features:** • Output Steam Capacity: 15-20 kg/ hr • Working Pressure: 2-3 bar • Power: 8.74 kw • Voltage: Three Phase 415V/50 Hz • Water level indicator and alarm system available • Automatic input water feeding system (ball value shall be automatically open when water level down and closed after water level reached) • Steam flow hose 1 m included • Input water storage 50 L cane- 1 No • Suitable electrical panel board with voltage switch control like MCB.	

	Electricals • MCC Panel board with power cables and control cables • Panel board, switches and cables – make and wire gauge should be mentioned. • Electrification Material like Cables, Cable trays, Clamps, Earthing, G.I strips necessary for power wiring from Panel to respective Motors. • Earthing, consisting of earth pits, running of earth wire etc for MCC panel and individual motor **Erection, Commissioning and trial run** • Complete erection of machines, drives, structure and electrical till MCCpanel. • Commissioning of total system toproduction. • Minimum two months training to locally appointed staff for operating extruded feed manufacturing unit. Training should be free ofcost. • Annual Maintenance charges should be free at least for first one year from the date of commissioning of feed plant. • Any service work should be attended within 48hours. • Guarantee and Warranty period should bespecified. • Spare parts with price list should besubmitted. **Structure** • Suitable structural design and quantities should be designed for standard production plant. • Stability certificate should be provided after erection. • Structural drawing for erection of machineries including platforms and ladder should be given. **Other terms and conditions:** • Machine should be built to comply with GMP & safety requirements. • Adequate spares required for one year should be mentioned. • Any expenditure incurred during erection and commissioning of the plant should also be included in the rate quoted.	

	• Company should have more than 5 years in the business. • Company should have the annual turnover more than Rs. 75 lakhs. • Company should provide the reference certificate from state/quasi government/ public sector or other reputed company. • One year warranty must be provided for the plant. • Installation and commission of plant is required at delivery location. • Company should show final product of required quality and quantity specified for issuance of completion certificate and payment.	